Calculations for Depth

$$v_t = \sqrt{\frac{2 \times (W - F_b)}{\rho \times C_d \times A}}$$

Where:

- W = Weight of the object in air ($m \times g$, where m is mass and g is gravity)
- F_b = Buoyant force (the weight of the water displaced by the object)
- ρ = Density of water (approx. $1000\ \text{kg/m}^3$ for fresh water)
- C_d = Drag coefficient (based on the object's shape)
- A = Projected surface area of the object

$$d = v \times t$$

Where:

- d = Depth (in meters)
- v = Sinking velocity (in meters per second, m/s)
- t = Total time it took to sink (in seconds)